

spaces we can replace the L^∞ -norm with the L^2 -norm. The application of Proposition 4.4.3 completes the proof. $\square$

4.5 FURTHER REMARKS AND CONJECTURES

In this Section we collect some thoughts about directions this work could be continued. We start by mentioning a more restricted class than that considered until now in this Chapter.

The Gevrey spaces we have considered so far are also known as *Gevrey–Roumieu* spaces. However, there is another class known as the *Gevrey–Beurling* spaces, which impose stricter conditions on the constants and are subspaces of the Gevrey–Roumieu spaces. We can state the adaptation, to the sub-Laplacians, of the definition of this second class.

Definition 4.5.1 ($\mathcal{L}$ -Gevrey–Beurling spaces, L^∞). Let $s > 0$. Let M be a manifold and $\mathbf{X} = \{X_1, \dots, X_r\}$ be a Hörmander system of vector fields. The sub-Laplacian Gevrey–Beurling space $\gamma_{\mathbf{X}, L^\infty}^{(s)}(M)$ of order s on a manifold M is the space of all functions $\phi \in \mathcal{C}^\infty(M)$ for which for every compact set $K \subset M$ and for any constant $A > 0$ there exists $C_A > 0$ such that for every $\alpha \in \mathbb{N}_0^r$ we have

$$|\partial^\alpha \phi(x)| \leq C_A A^{|\alpha|} (\alpha!)^s, \quad \text{for all } x \in K, \quad (4.5.1)$$

where $\partial^\alpha = Y_1 \dots Y_{|\alpha|}$, with $Y_j \in \{X_1, \dots, X_r\}$ for every $j = 1, \dots, |\alpha|$ and $\sum_{Y_j = X_k} 1 = \alpha_k$ for every $k = 1, \dots, r$.

It follows from the definition that $\gamma_{\mathbf{X}, L^\infty}^{(s)}(M) \subset \gamma_{\mathbf{X}, L^\infty}^s(M) \subset \mathcal{C}^\infty(M)$. As with the Gevrey–Roumieu spaces, we can define the L^2 -version as follows:

Definition 4.5.2 ($\mathcal{L}$ -Gevrey–Beurling spaces, L^2). The L^2 -version of the sub-Laplacian Gevrey–Beurling space, $\gamma_{\mathbf{X}, L^2}^{(s)}(M)$, of order s on M endowed with a measure μ , is the space of all functions $\phi \in \mathcal{C}^\infty(M)$ such that for any $A > 0$ there exists $C_A > 0$ such that for every $\alpha \in \mathbb{N}_0^r$ we have

$$\|\partial^\alpha \phi\|_{L^2(M)} \leq C_A A^{|\alpha|} (\alpha!)^s, \quad (4.5.2)$$

where ∂^α are as for (4.5.1).

Similarly to Proposition 4.2.3, under its assumptions, we have the inclusions

$$\gamma_{\mathbf{X},L^\infty}^{(s)}(M) \cap \mathcal{C}_0^\infty(M) \subset \gamma_{\mathbf{X},L^2}^{(s)}(M) \subset \gamma_{\mathbf{X},L^\infty}^{(s)}(M). \quad (4.5.3)$$

Remark 4.5.3. These inclusions mean that every result we have proved in this Chapter holds if one replaces the Gevrey–Roumieu space with its Gevrey–Beurling equivalent, so the latter may not seem particularly interesting. However their dual spaces, i.e., the spaces of continuous linear functionals on the respective original Gevrey spaces, satisfy the reverse inclusion. In light of the conjectures we will shortly formulate concerning these duals, this dual inclusion makes the Gevrey–Beurling spaces worthy of investigation in this topic.

4.5.1 HINTS ON ULTRADISTRIBUTIONS

As just mentioned, after introducing the sub-Laplacian Gevrey functions, we can also consider the spaces of linear functionals defined on the Gevrey classes. These spaces provide a bigger environment, comparable to that offered by the ‘classical’ distributions, where we might, for example, look for solutions of Cauchy problems. In this Subsection we conjecture theorems basing our claims on previous work on the compact case, e.g., [24]. It is our intention to verify thoroughly the details of the proofs in the near future.

Definition 4.5.4 (Ultradistributions). Let $s \geq 1$ and let M be a manifold. The spaces of continuous linear functionals on $\gamma_{\mathbf{X},L^2}^s(M)$ and $\gamma_{\mathbf{X},L^2}^{(s)}(M)$ are called the *spaces of ultradistributions* and are denoted respectively by

$$(\gamma_{\mathbf{X},L^2}^s)'(M) \quad \text{and} \quad (\gamma_{\mathbf{X},L^2}^{(s)})'(M).$$

Clearly $\mathcal{D}'(M) \subset (\gamma_{\mathbf{X},L^2}^s)'(M) \subset (\gamma_{\mathbf{X},L^2}^{(s)})'(M)$.

We refer to [24] or [25] for a more detailed discussion of such spaces. Combining our characterization for Gevrey spaces on $SU(2)$ and on the Heisenberg group with this work from Dasgupta and Ruzhansky on ultradistributions on compact groups

and manifolds, we expect to be able to characterise the corresponding spaces of (sub-Laplacian) ultradistributions in the settings of $SU(2)$ and of the Heisenberg group.

Since we can not talk about pointwise estimates for derivatives in the setting of distributions, it is most effective to aim at characterisations of these spaces on their Fourier transform side. By duality, the Fourier transform has an extension that acts on ultradistributions.

We follow the notation of Subsection 4.3.3 and Section 4.4, where sub-Laplacian Gevrey classes have been described on the Fourier transform side in the case of $SU(2)$ and the Heisenberg group, respectively. In particular, looking at the case of $SU(2)$, we recall the symbol of the sub-Laplacian from (4.3.4), so that in the theorem below, for every $l \in \frac{1}{2}\mathbb{N}_0$ being half-integer, $\widehat{\mathcal{L}}(l)$ is the diagonal matrix in $\mathbb{C}^{(2l+1) \times (2l+1)}$ with entries

$$\widehat{\mathcal{L}}(l)_{m,n} = (l(l+1) - m^2)\delta_{m,n}, \quad -l \leq m, n \leq l. \quad (4.5.4)$$

Conjecture 4.5.5. *Let $1 \leq s < \infty$. We have $u \in (\gamma_{\mathbf{X}, L^2}^s(\mathrm{SU}(2)))'$ if and only if for every $B > 0$ there exists $K_B > 0$ such that we have*

$$\|e^{-B\widehat{\mathcal{L}}(l)^{\frac{1}{2s}}}\widehat{u}(l)\|_{\mathrm{HS}} \leq K_B \text{ for all } l \in \frac{1}{2}\mathbb{N}_0. \quad (4.5.5)$$

We also have $v \in (\gamma_{\mathbf{X}, L^2}^{(s)}(\mathrm{SU}(2)))'$ if and only if there exist constants $B > 0$ and $K > 0$ such that

$$\|e^{B\widehat{\mathcal{L}}(l)^{\frac{1}{2s}}}\widehat{v}(l)\|_{\mathrm{HS}} \leq K \text{ for all } l \in \frac{1}{2}\mathbb{N}_0. \quad (4.5.6)$$

We expect the proof of Conjecture 4.5.5 to follow analogously to the proof of [24, Theorem 2.5] based on the characterisation from Corollary 4.3.6.

We recall that on the Heisenberg group the symbol of the sub-Laplacian $\mathcal{L}$ coincides with its infinitesimal representation $\pi_\lambda(\mathcal{L})$ given by (4.4.8). We also recall the Fourier transform

$$\widehat{f}(\lambda) = \int_{\mathbb{H}_n} f(x)\pi_\lambda(x)^* dx,$$

which can be extended to ultradistributions. The Plancherel formula takes the